Spectral Quaternion Lattice Inversion & Rational Field Projection

Algorithmic Architecture, Exact Diophantine Resolution over Q, and Sequential Function Reference (N = 100)

1. Concise Essay on the Algorithm

The **Spectral Quaternion Lattice Inversion & Rational Field Projection** algorithm establishes a deterministic computational bridge between analytic number theory, four-dimensional quaternion arithmetic, rational projective geometry, and random matrix universality. Given the first N non-trivial zeros $\rho_n = 1/2 + i\gamma_n$ of the Riemann zeta function $\zeta(s)$ and the corresponding sequence of prime numbers p_n (scaled from the 50-prime base case up to $N = 100$, $p_{100} = 541$), the algorithm maps spectral phase angles onto discrete Lipschitz quaternion shells and lifts every converged solution into exact rational coordinates in **Q**.

Stage I: Spectral Phase Projection & Determinant Initialization. For each prime-zero pair (p_n, γ_n), the algorithm computes the oscillatory Riemann phase angle $\theta_n = (\gamma_n \ln p_n) \bmod 2\pi \in [0, 2\pi)$. It then projects this circular phase onto the bounded integer determinant interval $[-\lfloor p_n/2 \rfloor, \lfloor p_n/2 \rfloor]$ via $D_0 = \text{round}((p_n/2) \cos \theta_n)$. In parallel, a seeded uniform random phase baseline ($\theta_{rand} \sim U(0, 2\pi)$) is initialized to test whether zeta-zero phases exhibit preferential alignment with integer quaternion lattices.

Stage II: Exact O(√p) Diophantine Lattice Resolution. Rather than performing an $O(p^2)$ four-dimensional brute-force search, the algorithm resolves the coupled Diophantine system $a^2 + b^2 + c^2 + d^2 = p$ and $ad - bc = D$ in $O(\sqrt{p})$ time. Using the algebraic identities $(a + d)^2 + (b - c)^2 = p + 2D$ and $(a - d)^2 + (b + c)^2 = p - 2D$, two-pointer scans find all two-square decompositions of $p + 2D$ and $p - 2D$, testing sign and parity combinations to recover $q = (a, b, c, d) \in Z^4$. If D_0 admits no integer representation, an expanding radius search tests $D_0 \pm k$ ($k = 1, 2, \ldots, \text{max_steps}$) until convergence.

Stage III: Exact Rational Field (Q) Lift & Invariant Verification. Once $q = (a, b, c, d)$ is found on the hypersphere $S^3(\sqrt{p})$, exact GCD-reduced fraction arithmetic lifts the tuple into **Q** without floating-point rounding. First, the normalized rational determinant $D_{conv}/p \in Q$ captures the quadratic residue parity law (even numerators for $p \equiv 1 \bmod 4$; odd numerators for $p \equiv 3 \bmod 4$). Second, quaternion squaring $U = q^2/p = ((a^2-b^2-c^2-d^2)/p, 2ab/p, 2ac/p, 2ad/p)$ maps every prime shell onto the true rational unit 3-sphere $S^3(Q)$ with exact norm $\|U\|^2 = 1/1$. Third, stereographic projection from the north pole maps (a, b, c, d) onto $(x, y, z) = (a/(p-d), b/(p-d), c/(p-d)) \in Q^3$, verified via exact cross-multiplication against $x^2 + y^2 + z^2 = (p - d^2)/(p - d)^2$.

Stage IV: Monte Carlo Hypothesis Testing & GUE Semicircle Unfolding. Across 500 Monte Carlo phase trials at $N = 100$, Riemann phases achieve 23/100 (23.00%) exact matches (mean 1.56 steps) versus 25.84 ± 4.00 (25.84%) for uniform random phases ($Z = -0.71$, $p = 0.7904$), confirming statistical equivalence to circular phase sampling. Finally, an optional spectral verification module diagonalizes a complex Hermitian GUE matrix and unfolds bulk eigenvalues via the exact Wigner semicircle CDF $F(\lambda)$, restoring unit local spacing density and the theoretical ~40.2% peak concentration in $0.75 < s < 1.25$.

Core Algebraic Pipeline:
1. Phase & Initial Det: $\theta = (\gamma \ln p) \bmod 2\pi$, $D_0 = \text{round}((p/2)\cos\theta)$
2. Coupled Lattice: $a^2 + b^2 + c^2 + d^2 = p$, $ad - bc = D_{conv}$ (via $p \pm 2D = x_{1,2}^2 + y_{1,2}^2$)
3. Exact Q Projections: $\text{Det}_Q = D_{conv}/p$, $U = q^2/p \in S^3(Q)$ ($||U||^2 = 1$), $\text{Stereo}_{Q^3} = (a, b, c)/(p - d)$

2. Sequential Execution Order of All Functions

The table below lists all 17 functions in the C++17 codebase in their exact sequential call and execution order from program entry to completion:

#	Function Name	Source File	Sequential Role & Computational Action
01	`main()`	`main.cpp`	Entry point orchestrating CLI parsing, zero loading, prime generation, node resolution, invariant checks, CSV export, Monte Carlo testing, and optional GUE verification.
02	`parse_cli()`	`main.cpp`	Parses command-line flags (`--count` , `--max-steps` , `--mc-trials` , `--seed` , `--gammas-file` , `--output` , `--verify-gue` , `--quiet`) into `RunConfig` .
03	`print_usage()`	`main.cpp`	Outputs CLI flag documentation and exits cleanly if `--help` or invalid arguments are supplied.
04	`load_gammas_from_file()`	`io_and_montecarlo.hpp`	Parses external LMFDB/Odlyzko zero files when `--gammas-file` is passed; otherwise `main()` slices the embedded 100-zero table `DEFAULT_RIEMANN_GAMMAS` .
05	`generate_primes()`	`lattice_solver.hpp`	Runs a dynamically sized Sieve of Eratosthenes using Rosser–Schoenfeld bounds to produce the first N primes ($p_1 = 2$ to $p_{100} = 541$).
06	`resolve_from_initial_d()`	`lattice_solver.hpp`	Tests D_0 for an exact lattice match (0 steps) or searches outward $D_0 \pm$ step up to `max_steps` for both Riemann and random baseline phases.
07	`solve_lattice()`	`lattice_solver.hpp`	Solves $a^2+b^2+c^2+d^2 = p$ and $ad-bc = D$ in $O(\sqrt{p})$ time by combining two-square decompositions of $p + 2D$ and $p - 2D$ across sign and parity combinations.
08	`solve_two_squares_all()`	`lattice_solver.hpp`	Uses a converging two-pointer scan from 0 to $\lfloor\sqrt{\text{target}}\rfloor$ to enumerate all integer pairs (X, Y) satisfying $X^2 + Y^2 = \text{target}$.

#	Function Name	Source File	Sequential Role & Computational Action
09	`lift_to_rationals()`	`rational_projection.hpp`	Maps converged integer quaternion q = (a, b, c, d) into reduced rational numbers: D_{conv}/p, unit quaternion norm $\|\|q^2/p\|\|^2 = 1$, and stereographic Q^3 coordinates.
10	`Rational::Rational()`	`spectral_inversion.h`	Constructs canonical reduced fractions num/den in Q by normalizing signs and dividing by `std::gcd(|n|, d)`.
11	`verify_node_invariants()`	`main.cpp`	Validates integer invariants ($a^2+b^2+c^2+d^2 = p$ and $ad-bc = D_{conv}$) and delegates rational field checks for every solved node.
12	`verify_rational_invariants()`	`rational_projection.hpp`	Verifies exact rational unit norm (1/1) on $S^3(Q)$ and exact integer cross-multiplied stereographic identity $a^2 + b^2 + c^2 = p - d^2$.
13	`export_side_by_side_csv()`	`io_and_montecarlo.hpp`	Serializes all *N* nodes—including Z^4 tuples, Q rational projections, search steps, and step differences—to `spectral_comparison_100.csv`.
14	`run_monte_carlo_comparison()`	`io_and_montecarlo.hpp`	Executes `mc_trials` (500) uniform random phase simulations across all *N* primes to compute baseline mean, standard deviation, Z-score, and empirical p-value.
15	`verify_gue_semicircle_unfolding()`	`gue_unfolding.hpp`	When `--verify-gue` is enabled, constructs an $N_{GUE} \times N_{GUE}$ complex Hermitian matrix, unfolds bulk eigenvalues via Wigner's semicircle CDF, and measures spacing statistics.
16	`compute_hermitian_eigenvalues()`	`gue_unfolding.hpp`	Reduces the complex Hermitian matrix to real symmetric tridiagonal form via Householder reflections and diagonalizes via Wilkinson-shifted implicit QL iteration.
17	`Rational::to_double() / str()`	`spectral_inversion.h`	Helper formatters invoked during table/CSV output to render reduced rational fractions as strings (`num/den`) or floating-point values.

Verified N = 100 Benchmark Summary: Evaluated 100 prime–zero pairs (p_{max} = 541, max search radius = 6 steps). Riemann exact matches: **23/100 (23.00%)**, mean steps: **1.56**; 500-trial Monte Carlo baseline: **25.84 ± 4.00 (25.84%)**, Z-score: **−0.71**, one-sided p-value: **0.7904**. All 100 integer and rational field invariants verified with zero errors.